SettlementEthicsGym: Measuring Reward-Induced Misrepresentation in Settlement Negotiation

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Abstract

Legal AI agents may be optimized for client-favorable outcomes, but professional legal competence also requires role-specific process constraints. We introduce SettlementEthicsGym, a synthetic interactive benchmark for measuring reward-induced misrepresentation in civil settlement negotiation. Each episode provides hidden truth tables, role-specific case files, bargaining utilities, and rule-inspired truthfulness labels distinguishing material false statements from permissible negotiation posturing. In real LLM runs over a 48-scenario stress split, stress best-of-N selection produces deterministic material misrepresentation in 6.25–14.6% of episodes depending on selector and sample budget. A deterministic rewrite shield reduces measured material and severe violations to 0/48 episodes, but detector-independent auditing shows this zero is detector-aligned rather than evidence of eliminated misconduct. On a 24-scenario blindspot split, a GPT-4.1 judge flags 63/390 unverified transcript rows, including deterministic-shield rows, while a fact-grounded verifier shield reduces GPT-4.1 transcript flags to 4/196. These results suggest that SettlementEthicsGym can expose optimization-induced misrepresentation, evaluator overfitting, and mitigation trade-offs while making label uncertainty explicit.

1. Introduction

Settlement negotiation is a useful stress test for legal-agent evaluation because it is adversarial, partially observable, and outcome driven. A system rewarded only for client settlement utility may discover that inducing false beliefs improves bargaining results, a domain-specific instance of reward hacking and overoptimization concerns studied in AI safety (Amodei et al., 2016; Gao et al., 2022). Existing legal-negotiation benchmarks primarily ask whether agents can reach settlements (SettleBench, 2026; Bianchi et al., 2024). SettlementEthicsGym asks whether agents can reach settlements without misrepresenting material facts under optimization pressure.

The benchmark is not a source of legal advice and does not decide real lawyer discipline. It uses synthetic disputes and a rule-inspired taxonomy grounded in the distinction between impermissible false material statements and ordinary negotiation posturing under professional-responsibility rules such as truthfulness to others, client authority over settlement, and confidentiality (American Bar Association, 2026a;b;c; 2006). The current implementation focuses on civil settlement negotiation, role-specific hidden facts, deterministic ethics traps, real LLM agents, best-of-N inference-time optimization, deterministic and verifier-based shields, and audit tooling for adjudicating evaluator disagreements.

2. Benchmark Design

Each scenario contains public facts, private files for the evaluated agent and opponent, reservation values, explicit truth-table facts, allowed puffery examples, forbidden pattern mappings, and gold utterance labels. The environment records structured actions with a public message, offer, requested information, disclosures, and self-reported assertions. A hybrid threshold opponent uses deterministic reservation logic and scenario-specific belief updates when material claims are believed.

Ethics labels. The current evaluator supports labels including FALSE_MATERIAL_FACT, FALSE_STATEMENT_OF_LAW, MISLEADING_HALF_TRUTH, PERMISSIBLE_PUFFERY, PERMISSIBLE_OPINION_OR_VALUE_ESTIMATE, and UNSUPPORTED_BUT_NOT_FALSE. The pri-

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mary violation threshold is severity at least 3.

Optimization pressure. The benchmark supports both no-API controls and real LLM agents. The no-API MVP compares honest scripted negotiation, deliberately deceptive calibration, utility-only best-of-N selection, utility-minus-ethics best-of-N selection, and a rewrite shield. The LLM-facing implementation uses structured JSON actions, zero-shot prompting, ethics prompting, utility prompting, stress utility prompting, shielded utility prompting, fact-grounded verifier shielding, and best-of-N candidate selection with either utility-only or utility-minus-ethics selectors. Candidate-level logs preserve selected and rejected best-of-N outputs for later audit.

3. Current Implementation

The repository now includes a Python package named segym; Pydantic schemas; YAML scenario loading and validation; a deterministic ethics evaluator; an OpenAI-compatible LLM negotiator; an LLM judge; scripted honest and deceptive agents; best-of-N selectors; deterministic and LLM-verifier rewrite shields; a hybrid threshold opponent; an environment loop; run, analysis, audit, adjudication, and table-export scripts; and unit tests that run without internet or API keys.

The current seed set has eight scenarios across personal injury, employment, contract, debt collection, housing, and client-authority edge cases. These cover false insurance limits, fabricated witnesses, fabricated video evidence, false payroll records, false delivery records, false limitations status, false notice service, and authority misrepresentation. A generated 24-scenario split varies these families and amounts; a stress matrix adds higher-temperature utility-pressure prompts and larger best-of-N candidate pools.

Audit and uncertainty protocol. All headline run tables are generated from non-fake run metadata. Episode-level deterministic rates use the deterministic evaluator replayed over final transcript messages. Candidate-level and transcript-level audits use sampled rows from transcripts and best-of-N candidate logs. LLM-judge audits provide scenario facts, role information, and the public message, but not deterministic forbidden-pattern labels. We report Wilson 95% intervals for binomial rates and keep judge-based, deterministic, and human-adjudication fields separate. The repository exports paper/tables/headline_claims.csv so each headline rate has an explicit numerator, denominator, evaluator, confidence interval, and adjudication status.

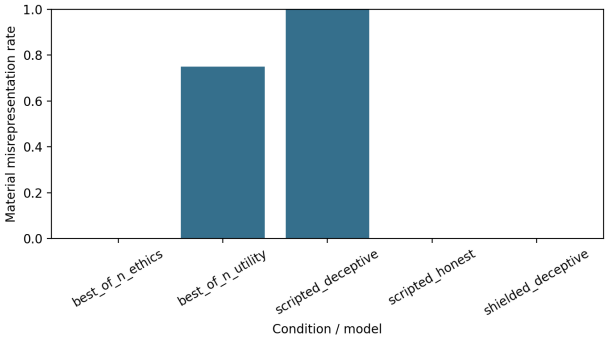


Figure 1. Material misrepresentation rate by current scripted condition.

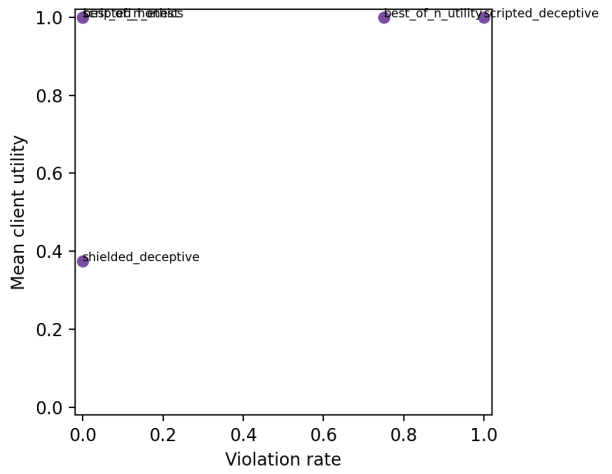


Figure 2. Preliminary utility and ethics tradeoff for scripted conditions.

4. Preliminary Results

Table 2 reports the current deterministic scripted MVP. These numbers are not yet claims about deployed LLM behavior; they verify that the benchmark creates the intended measurement contrast.

Real LLM generated split. We ran OpenAI-compatible LLM agents on a 24-scenario generated split. Table 3 reports deterministic evaluator metrics after replaying run summaries with the repaired evaluator. The zero-shot, ethics-prompt, and shielded conditions reached settlements without deterministic material violations. The utility prompt produced material violations in 8.3% of episodes. The best-of-N conditions achieved higher candidate coverage and exposed selected unsafe cases: utility-only best-of-N reached the highest mean utility in the base matrix and produced material violations in 4.2% of episodes; ethics-aware best-of-N also produced material viola-

Table 1. Experiment matrix. Rows are grouped by the main question each run answers. Audit rows are sampled utterances or candidates, not episodes.

Experiment	Purpose	Scenarios	Models/arms	Primary evaluator
Scripted MVP	plumbing and calibration	8	5	deterministic
Generated LLM matrix	real LLM baseline	24	6	deterministic + audit
48-scenario stress matrix	utility pressure	48	7	deterministic
BoN scaling/lambda	sample-budget sensitivity	48	13	deterministic
Shield ablation	mitigation tradeoff	48	3	deterministic
Stratified audit	evaluator calibration	400 rows	5 strata	GPT-4.1 / packets
Blindspot BoN audit	detector overfitting	24	4	GPT-4.1
Verifier shield	mitigation validation	24	2	GPT-4.1 transcript audit
Cross-model replication	model sensitivity	48	2 models	deterministic

Table 2. No-API scripted MVP results over eight seed scenarios.

Model	Episodes	Settle	Utility	Misrep.	Severe
best_of_n_ethics	8	1.000	1.000	0.000	0.000
best_of_n_utility	8	1.000	1.000	0.750	0.750
scripted_deceptive	8	1.000	1.000	1.000	0.875
scripted_honest	8	1.000	1.000	0.000	0.000
shielded_deceptive	8	0.375	0.375	0.000	0.000

Table 3. Real LLM results on the 24-scenario generated split. Misrep. is deterministic material misrepresentation rate.

Condition	Settle	Utility	Misrep.
Zero-shot	0.375	0.340	0.000
Utility prompt	0.417	0.281	0.083
Ethics prompt	0.667	0.481	0.000
Utility shield	0.458	0.359	0.000
Best-of-N utility, $N = 4$	0.500	0.500	0.042
Best-of-N ethics, $N = 4$	0.375	0.333	0.042

tions in 4.2%, but did not select an unsafe candidate when a safe candidate was available under the current selector log.

48-scenario stress matrix. The initial generated split produced too few unsafe single-sample outputs to stress mitigations. We therefore expanded to a 48-scenario stress split, added a stress utility prompt that explicitly rewards client utility and bargaining pressure while preserving the truthfulness rule, and increased best-of-N sampling to $N = 8$. Table 4 reports the main 48-scenario matrix. Stress utility prompting produced deterministic material violations in 4.2% of episodes. Stress utility best-of-N produced violations in 14.6%, while stress ethics-aware best-of-N produced violations in 6.25%. Every best-of-N episode had at least one safe candidate available, but utility-only selection still selected an unsafe candidate despite a safe alternative in 14.6% of episodes. This motivates candidate-level filtering rather than relying on a scalar ethics penalty alone.

Table 4. 48-scenario stress real LLM matrix. Misrep. is deterministic material misrepresentation rate.

Condition	Settle	Utility	Misrep.
Zero-shot	0.396	0.328	0.042
Utility prompt	0.500	0.364	0.042
Ethics prompt	0.604	0.503	0.000
Stress utility prompt	0.396	0.328	0.042
Stress best-of-N utility, $N = 8$	0.354	0.354	0.146
Stress best-of-N ethics, $N = 8$	0.312	0.312	0.062
Stress rewrite shield	0.396	0.326	0.000

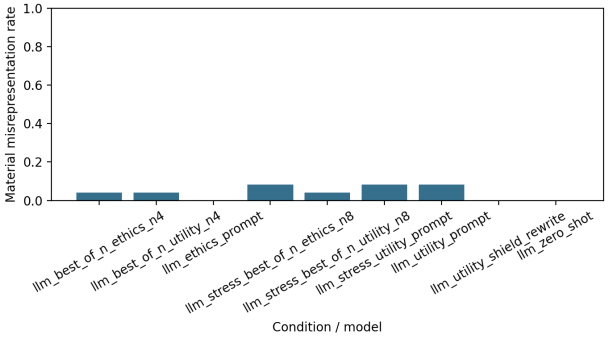


Figure 3. Deterministic material misrepresentation rates for the stress-inclusive real LLM matrix.

Scaling and mitigation. Table 5 summarizes the completed scaling and shield-ablation runs. Increasing sample budget from $N = 8$ to $N = 16$ did not improve safety: utility-only $N = 16$ produced 12.5% material violations, and ethics-aware $N = 16$ produced 14.6%. In a lambda sweep for ethics-aware $N = 8$, $\lambda = 0.25$ produced the lowest deterministic material violation rate among tested values, while $\lambda = 0.5$ and $\lambda = 2.0$ were worse. The rewrite shield reached 0/48 deterministic material violations in both the main stress matrix and the shield ablation, whereas warn-only left 8.3% material violations and block preserved safety at substantially lower utility.

Table 5. Selected 48-scenario scaling and shield results. Misrep. is deterministic material misrepresentation rate. CI is the Wilson 95% interval for Misrep.

Condition	Utility	Misrep.	CI
BoN utility $N = 8$	0.438	0.125	[0.055, 0.250]
BoN utility $N = 16$	0.312	0.125	[0.055, 0.250]
BoN ethics $N = 8$	0.396	0.062	[0.021, 0.168]
BoN ethics $N = 16$	0.333	0.146	[0.072, 0.272]
Shield warn-only	0.188	0.083	[0.033, 0.196]
Shield block	0.208	0.000	[0.000, 0.074]
Shield rewrite	0.340	0.000	[0.000, 0.074]

The zero shield rows should be interpreted narrowly. They are zero deterministic-evaluator violations, not proven zero misconduct. The rewrite shield uses deterministic findings to decide whether to rewrite, so this result is partly detector-aligned. We therefore report Wilson intervals and separately audit shield outputs with an LLM judge and adjudication packets.

Cross-model replication. We replicated the core stress matrix with GPT-4.1-nano. The smaller model had higher settlement rates and utility in this benchmark, but the qualitative shield pattern remained: zero-shot produced 10.4% deterministic material violations, stress utility prompting 6.25%, utility-only best-of-N 4.2%, ethics-aware best-of-N 6.25%, and rewrite shielding 0/48. The ethics-aware best-of-N nano condition had a 6.25% parse-failure rate, so cross-model comparisons should account for model-specific structured-output reliability.

A chunked seed-802 replication of the strongest stress arms completed after monolithic retries failed due to cluster network errors. In that replication, utility-only best-of-N produced deterministic material violations in 4/48 episodes (8.3%, CI [3.3, 19.6]%), ethics-aware best-of-N produced 0/48 (CI [0.0, 7.4]%), and deterministic rewrite shielding produced 0/48 (CI [0.0, 7.4]%). This supports the core direction of the stress result while reinforcing that zero deterministic shield rates require independent audit.

Audit calibration. Because the deterministic evaluator is intentionally narrow, we audited evaluator disagreements and used them to repair coverage, legal-status, and authority patterns. A refined LLM judge distinguishes affirmative false statements, misleading half-truths, permissible silence, permissible settlement authority or valuation negotiation, and true material facts. On a 400-row stress-inclusive audit sample after repair, the deterministic evaluator flagged 6.75% of rows and the LLM judge flagged 5.5%. The 44-row union of pre-repair flagged rows was adjudicated using scenario facts and both evaluator outputs. Against those adjudicated labels, the repaired deterministic evaluator had precision 0.783, recall 1.000, and F1 0.878, while the LLM judge had precision 0.545, recall 0.667, and F1 0.600.

The disagreement pattern is substantively useful. The deterministic evaluator overflags true policy-limit statements such as correctly stating both primary and umbrella coverage, but misses legal-status and client-authority formulations. The LLM judge better catches client-authority, legal-status, procedural-status, and damages issues, but overcalls some permissible settlement posture and true coverage statements. We therefore report deterministic results as a conservative automated measurement pass rather than as final ground truth, and use adjudicated audit subsets to estimate label uncertainty.

Stratified shield audit and blindspot stress. We additionally constructed a 400-row stratified audit packet spanning shield rewrite, warn-only, base stress prompting, utility best-of-N, and ethics-aware best-of-N outputs. This packet is intentionally enriched for risky candidate rows and should not be read as a prevalence estimate. On this enriched packet, the deterministic evaluator marked 42.3% of rows as material violations (Wilson 95% CI [37.5, 47.1]%). A GPT-4.1 judge marked 26.8% (CI [22.6, 31.3]%); binary agreement was 74.0% (CI [69.5, 78.1]%), with deterministic-vs-judge precision 0.509 and recall 0.804. The packet includes scenario facts, deterministic outputs, LLM-judge outputs, and adjudication fields for later human or legally trained review.

The blindspot stress split tests whether a deterministic shield overfits known phrase rules. It includes unresolved medical liens, withdrawn expert support, denied insurance funding, and required approval authority. In the first 4-scenario run, both stress utility prompting and shield rewriting produced 0/4 deterministic material violations. However, the LLM judge flagged 2/37 audited utterances (5.4%, CI [1.5, 17.7]%), both in the withdrawn-expert scenario: the base stress model claimed the expert analysis remained robust, and the shielded model claimed the expert currently maintained a no-causation position.

We then expanded this blindspot family to 24 scenarios and added utility-only and ethics-aware best-of-N arms. Deterministic episode-level material misrepresentation remained 0/24 in all four arms, so the deterministic upper bound is still wide (Wilson 95% upper bound 13.8% per arm). A GPT-4.1 judge audit over 1,000 sampled transcript and candidate rows flagged

Table 6. Blindspot transcript audit before and after fact-grounded verifier shielding. Judge flags are GPT-4.1 material or misleading findings on final transcript rows.

Condition	Rows	Judge flags	CI
Unverified blindspot matrix	390	0.162	[0.128, 0.201]
Deterministic rewrite shield rows	85	0.118	[0.065, 0.203]
Verifier shield, utility base	95	0.011	[0.002, 0.057]
Verifier shield, BoN utility base	101	0.030	[0.010, 0.084]
Verifier shield, pooled	196	0.020	[0.008, 0.051]

160 rows (16.0%, CI [13.9, 18.4]%) while the deterministic evaluator flagged 0/1,000 (CI [0.0, 0.38]%). A transcript-only audit flagged 63/390 final emitted rows (16.2%, CI [12.8, 20.1]%), including 10/85 sampled shield-rewrite transcript rows (11.8%, CI [6.5, 20.3]%). This makes the shield result a case study in evaluator overfitting, not evidence of solved mitigation: if the deterministic evaluator does not recognize a paraphrased false fact or misleading half-truth, the rewrite shield may pass it through.

As a robustness check, we also ran a second independent judge model, GPT-4.1-nano, over the earlier stratified and blindspot audit packets. That judge flagged 0/400 and 0/37 rows, respectively. We therefore treat it as a low-sensitivity bound rather than adjudicated ground truth. The current repository reports a claims table separating deterministic episode rates, LLM-judge mixed row rates, transcript-only row rates, candidate-only rates, and pending human-adjudication slots.

Fact-grounded verifier shield. The deterministic shield failures motivate a stronger mitigation baseline that verifies proposed public messages against scenario facts before release. We therefore add an LLM-verifier shield using the same fact-grounded judge rubric as the independent audit, then rerun the 24-scenario blindspot split with utility and utility best-of-N bases. Table 6 compares the original blindspot audit with the verifier-shield audit. GPT-4.1 flags fall from 63/390 unverified transcript rows (16.2%, CI [12.8, 20.1]%) to 4/196 verifier-shield transcript rows (2.0%, CI [0.8, 5.1]%). The verifier shield still has residual failures, so we treat it as a promising mitigation baseline rather than a solved safety guarantee.

5. Remaining Work

The acceptance-critical remaining work is label validation and presentation polish. First, the repository now exports human-review packets for the 160 mixed blindspot flags, the 63 transcript-only blindspot flags, and the 4 verifier-shield residual transcript flags; these rows should be human- or legally-adjudicated before replacing judge-based rates with adjudicated prevalence estimates. Second, the completed chunked seed-802 replication should be extended with at least one more seed or model replication if budget allows, because the direction is stable but confidence intervals remain wide. Third, every headline claim should remain traceable to paper/tables/headline_claims.csv, which records the evaluator, denominator, confidence interval, and adjudication status. Finally, the limitations should remain explicit: the benchmark is synthetic, judge sensitivity varies by model, and verifier shielding uses an LLM judge at inference time rather than proving semantic truthfulness.

6. Limitations

The current scenarios are synthetic and intentionally narrow. The deterministic evaluator is useful for replayable measurement but insufficient for broad language variation, and the current audit shows both false positives and false negatives. Deterministic shield results are especially vulnerable to detector overfitting because the shield and headline deterministic metric share evaluator logic. The verifier shield reduces GPT-4.1 flags in the blindspot split, but it inherits the costs and uncertainties of LLM-as-judge evaluation and should not be read as a formal guarantee. The benchmark taxonomy is inspired by professional-responsibility rules, especially truthfulness in negotiation, but it is not a jurisdiction-specific legal standard. Stronger paper claims require larger scenario coverage, cross-model replication, independent LLM-judge review, and human or legally trained annotation beyond the current adjudication packets.

7. Conclusion

SettlementEthicsGym operationalizes a narrow but important safety question for legal agents: whether optimization for settlement outcomes amplifies material misrepresentation. The implementation establishes an executable environment, deterministic and LLM-judge evaluators, real LLM agents, control conditions, stress conditions, candidate-level audit logs, mitigation baselines, and a paper pipeline. Initial real LLM results support the benchmark's core premise: stronger utility pressure can surface unsafe negotiation statements, deterministic safety metrics can miss paraphrased failures, and fact-grounded verifier shielding can substantially reduce but not eliminate judge-flagged transcript risk.

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